

LLM 4-dars.

FFN $\Rightarrow$ Knowledge Engage xosil qiladi

RMSNorm $\Rightarrow$ Aynan bir qiymatni $\sqrt{\cdot}$ siqa
bölish kifoya qilar ekan

MHA $\rightarrow$ Har bir sozga shadan key va value beradi
GPE $\rightarrow$ Sozlarni ma'lum guruhlarqa bōlib ularga
key va value beradi.

Performance uchun kerakli jarayonlar:

- Hyperparameters
- Regularization
- Training stability

Hyperparameters: LR, Batch Size, Optimizer, Warm.Steps,
Weight-Decay

LR Scheduler: Natijaga yetganda LR ni oga pasaytirish
Vaqt o'tishi b/n qadam uzunligini
gauss grafik asosida pasaytirib borish

Global Batch Size: 1 ta matematik qadamda GPU lar jami bōlib qancha joykuni bir vaqtda xisoblash lajmi

Optimizer uses LR to update model weights and lower error during training.

Optimizer picks the path, and **LR** sets the step size.

Warmup Steps: model juda kichik LR bly boshlashi

Weight Decay: Overfittingni oldini dīsh jarayoni

Gradient Exploding: Gradient larni ortib ketishi

←

Yechimi LR warmup, **Gradient Clipping**.

LR warmup + Gradient Clipping $\approx$ Stability